SYSTEM ANALYSIS & DESIGN

# SmartListing Copilot

Multimodal Listing Quality Control for Fashion E-Commerce

## UI/UX Design Package · Member 5 deliverable

| | |
|---|---|
| **Version** | 1.2 (rules, integration contract and walkthrough pack added) |
| **Date** | 28 October 2026 |
| **Status** | Prepared for team review; stakeholder sign-off pending |
| **Deliverable** | Design package and interactive UI prototype (prototype.html) |

**Prototype boundary.** This is a proposed design and an interactive UI demonstration. It is not a deployed or model-connected application; all analysis results shown are hard-coded demonstration scenarios.

## Document control and source basis

This package was prepared from the SmartListing Copilot Requirements Documentation (revised 16 October 2026), the Functional Requirements Specification (FRS) PDF, and the revised-scope project proposal. No project repository source code or running application was available in the inspected files, so the included prototype is standalone.

| Item | Baseline / status |
|---|---|
| **Dataset** | Fashion Product Images Small is the sole MVP dataset context. Five EDA-selected categories are expected to be frozen by the team. |
| **Attributes** | Product type and color only. Size, material and price are out of scope. |
| **Description** | Seller-entered or synthetic test text. The dataset does not provide a description field. |
| **Suggestions** | Controlled/template suggestions; the user can Accept, Edit or Ignore. No automatic publishing. |
| **Approval** | Design walkthrough and stakeholder sign-off remain pending. |

### Revision history

| Version | Date | Change |
|---|---|---|
| 1.0 | 27 Oct 2026 | Initial design package and prototype. |

| Version | Date | Change |
|---|---|---|
| 1.1 | 28 Oct 2026 | Analysis screen redesigned (staged pipeline, progress bar, listing context). Review Results screen redesigned (summary banner, count tiles, evidence comparison, highlighted before/after suggestion). Screenshots of every state added; user flow diagram added. |
| 1.2 | 28 Oct 2026 | Verified against the FRS and proposal. Added Section 12 (category set and validation rules), Section 13 (pipeline integration contract) and Section 14 (walkthrough and sign-off pack). Prototype now shows the approved reason code, severity, inconsistency level and queue; category list updated to the proposal candidate set. FR/US/AC IDs confirmed. |

## Review and approval

To be completed after the design walkthrough.

| Name | Role | Date | Signature |
|---|---|---|---|
| | | | |
| | | | |

# 1. UI/UX scope and assumptions

The interface supports one listing-review journey: enter product details, upload one image, submit, inspect the image-supported and text-claimed product type and color, review any mismatch or uncertainty, then edit or finish. It is not a marketplace, inventory tool, order system, or publishing integration.

## Confirmed constraints

- Fashion Product Images Small is the sole MVP dataset context.
- Only product type and color are checked.
- Unknown attributes are not inferred; confidence is not verified correctness.
- Comparisons may identify consistent, contradictory or unclaimed supported attributes.
- Suggested corrections require user choice and must not auto-publish.

## Assumptions requiring confirmation

- The prototype requires image, title, description and seller-stated category because the FRS input table marks these as required; empty-text behavior is also noted as unresolved.
- Accepted image encodings, file-size limits and text-length limits are not specified in the FRS or the proposal and are not invented here (see Section 12.2).
- English and a fixed Arabic lexicon are referenced by the FRS; a full bilingual UI is not claimed in this prototype.
- All analysis outputs are hard-coded scenarios; no model, API, persistence or real score is connected.
- Prototype category options are the proposal's illustrative candidate set (Casual Shoes, Formal Shoes, Sports Shoes, Shirts, Handbags); the final five classes are confirmed after EDA (see Section 12.1).

# 2. User persona

| | |
|---|---|
| **Role** | Mariam Hassan, Marketplace Listing Specialist. Illustrative persona, not an interviewed participant. |
| **Context** | Prepares product titles and descriptions from product photos before listing submission to a separate publishing workflow. |
| **Goals** | Catch type/color inconsistencies, understand flags, correct text quickly, and retain final control. |
| **Pain points** | Repetitive manual checks, vague warnings, uncertainty presented as fact, and suggestions that change the intended meaning. |
| **Expectations** | Single-item workflow, clear comparison, plain-language evidence, editable suggestions, and safe retry. |
| **Evidence status** | Hypothesis inferred from requirements; validate with seller/catalog reviewer walkthroughs. |

# 3. User journey

| Stage | User action | Proposed system response | Friction and design response |
|---|---|---|---|
| **Open** | Opens the tool | Explains the task and the type/color scope | Avoid feature overload. |
| **Enter** | Uploads an image and enters title, description and category | Shows a preview and persistent labels | Field-level validation and required markers. |
| **Validate** | Submits | Checks required fields and image readability | Focus the first invalid field; preserve entered data. |
| **Analyze** | Waits | Shows the submitted image, a four-stage pipeline and a progress bar; disables duplicate submission | Only report stages actually executed in production. |
| **Review** | Reads outputs | Shows an outcome banner and counts, then separates image prediction from text claim; explains the finding and uncertainty | Do not use color alone to convey status. |
| **Decide** | Accepts, edits or ignores the suggestion | Shows before/after wording with the changed word highlighted | No automatic publishing. |
| **Finish** | Edits or finishes | Safe recovery from errors | Distinguish service failure from mismatch. |

# 4. User flow diagram

The diagram below shows the proposed behavior, not a verified implementation. The Mermaid source is provided separately as user_flow.mmd.

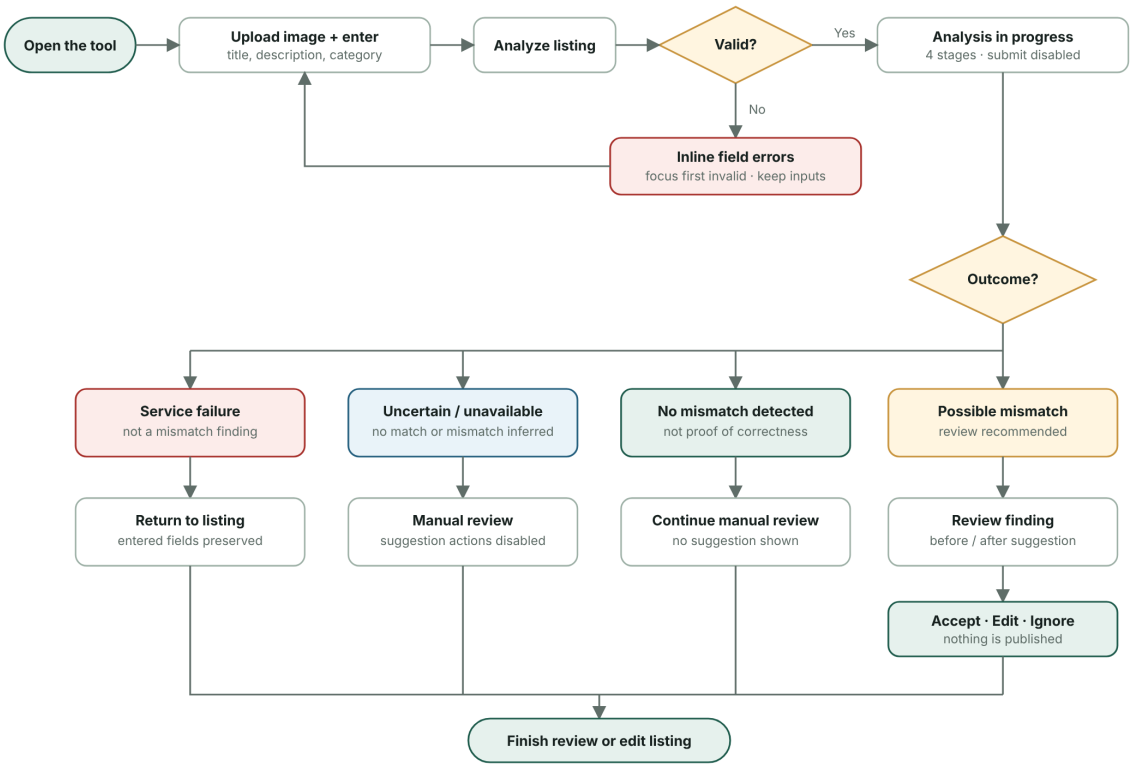



**Figure 1.** *Proposed user flow for a single listing review.*

# 5. Annotated low-fidelity wireframes

The wireframes are specified region by region. The high-fidelity screens in Section 6 show how each region is realized in the prototype.

## 5.1 Main / product input

| Region | Wireframe content | Purpose |
|---|---|---|
| **Header** | SmartListing Copilot and a "UI demo" badge | Brand and prototype boundary. |
| **Hero** | Listing quality control; type/color scope | Set expectations. |
| **Uploader** | Choose/drop image, then preview and filename | Confirm the selected asset. |
| **Fields** | Title*, description*, seller-stated category* | Explicitly labelled input. |
| **Validation** | Inline message below the invalid field | Actionable error recovery. |
| **Primary action** | Analyze listing | Submit after validation. |
| **Side guidance** | How it works and scope guardrails | Explain the supported workflow. |

## 5.2 Analysis in progress

| Region | Content | Purpose |
|---|---|---|
| **Context header** | Thumbnail of the submitted image with a scanning effect, title, and chips for the entered title and category | Confirm which listing is being analyzed. |
| **Progress bar** | Percentage and the current stage label | Show overall progress at a glance. |
| **Stage pipeline** | Inputs validated / reading the image / reading the listing text / comparing supported attributes, each marked Done, In progress or Waiting | Concise status in text, not color alone; only real stages in production. |
| **Submit control** | Disabled during pending analysis | Prevent duplicate submissions. |
| **Recovery** | Cancel demo returns to the form with entries preserved | Prototype affordance; production cancellation must be confirmed. |

## 5.3 Review results

| Region | Content | Purpose |
|---|---|---|
| **Summary banner** | Outcome icon, headline (Review recommended / No mismatch / Manual review) and one-line explanation | Outcome in text, icon and status label. |
| **Review details** | Chips for reason code, severity, inconsistency level and review queue | Show the approved rule outcomes (Section 12). |
| **Count tiles** | Attributes checked, consistent, needs review, uncertain | Quick quantitative overview. |
| **Submitted listing** | Image, filename, title, description, category | Preserve evidence and context beside the findings. |
| **Comparison** | Attribute / image-supported / text claim / status | Keep signals distinct. |

| Region | Content | Purpose |
|---|---|---|
| **Finding** | Evidence, explanation and approved severity | Make the flag explainable. |
| **Suggestion** | Original wording with the conflicting word highlighted, beside editable suggested wording | Transparent, human-controlled correction. |
| **Actions** | Accept / Edit / Ignore / Edit listing / Finish review | No automatic publishing. |

## 5.4 Error and empty states

| State | Behavior | Recovery |
|---|---|---|
| **Missing information** | Inline field-level messages | Keep inputs; focus the first invalid field. |
| **Unsupported/unreadable image** | Explain the image cannot be analyzed; exact format limits pending | Replace the image. |
| **Analysis failure** | Explain the operation failed, not that a mismatch was found | Retry or return to the form. |
| **No mismatch** | Say no mismatch was detected in supported attributes; not a guarantee | Continue manual review. |
| **Uncertain/unavailable** | Do not infer a match or mismatch | Manual review; suppress unsupported correction. |
| **Missing claim** | Label as "not stated", not contradictory | Offer only approved supported type/color templates. |

# 6. High-fidelity prototype

Open **prototype.html** in a modern browser. It is a standalone, responsive HTML/CSS/JavaScript demonstration with image upload and preview, drag-and-drop, validation, a staged simulated analysis, mismatch / no-mismatch / uncertain scenarios, an editable suggestion with Accept / Edit / Ignore controls, and a simulated failure state. No installation or server is required.

> **Demonstration data only.** All results are hard-coded. The black-sneaker image and the white-text contradiction are illustrative, not measured model performance. The "Fill with sample listing" link pre-loads this example so every screen can be reproduced quickly. Screenshots in this section were captured from the prototype in a desktop browser (1280 px) and a 390 px mobile viewport.

## 6.1 Product input

The input screen collects the image, title, description and seller-stated category. Required fields carry a textual asterisk and the right-hand panel explains the workflow and the scope guardrails.

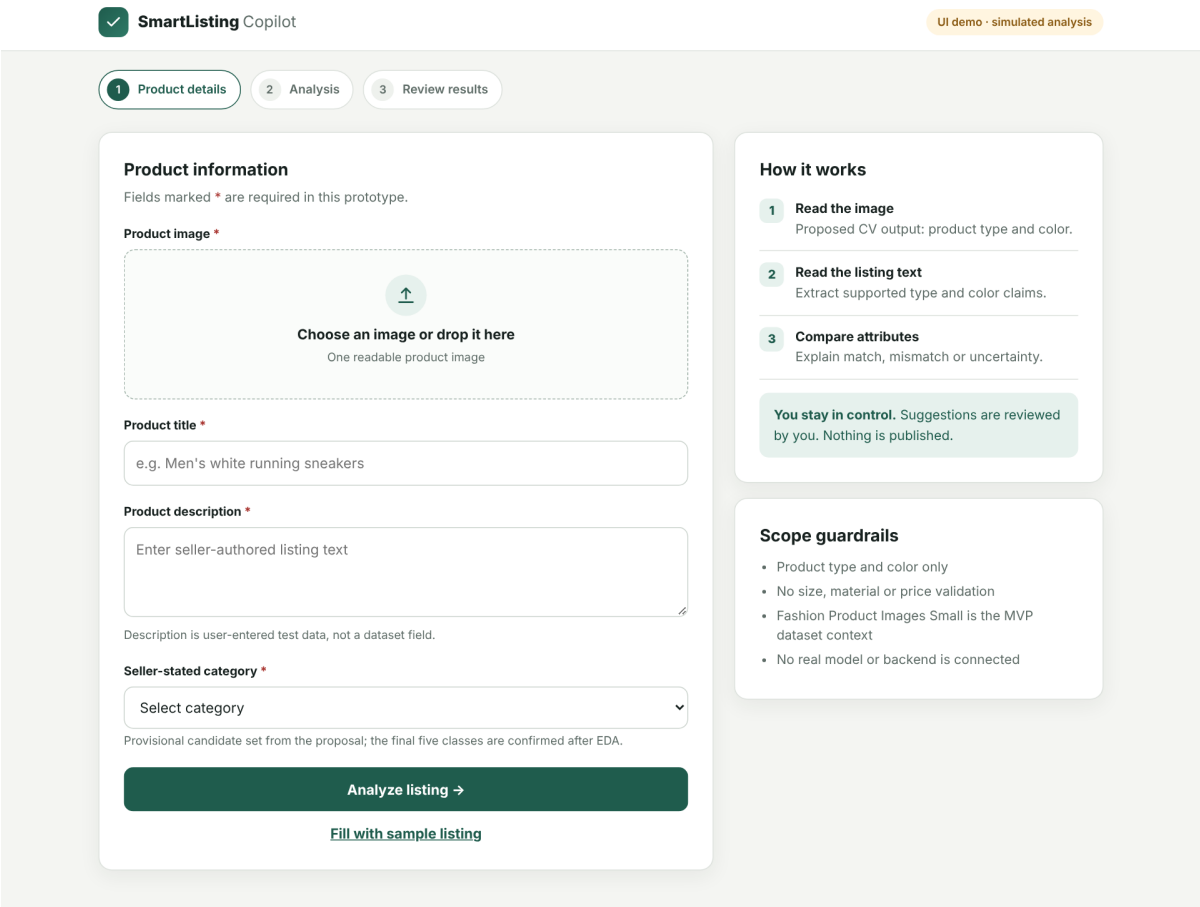



**Figure 2.** *Product input screen (empty state, desktop).*

## 6.2 Field-level validation

Submitting an incomplete form highlights each invalid field with an inline message, shows a summary count, and moves focus to the first invalid field. Entered data is preserved.

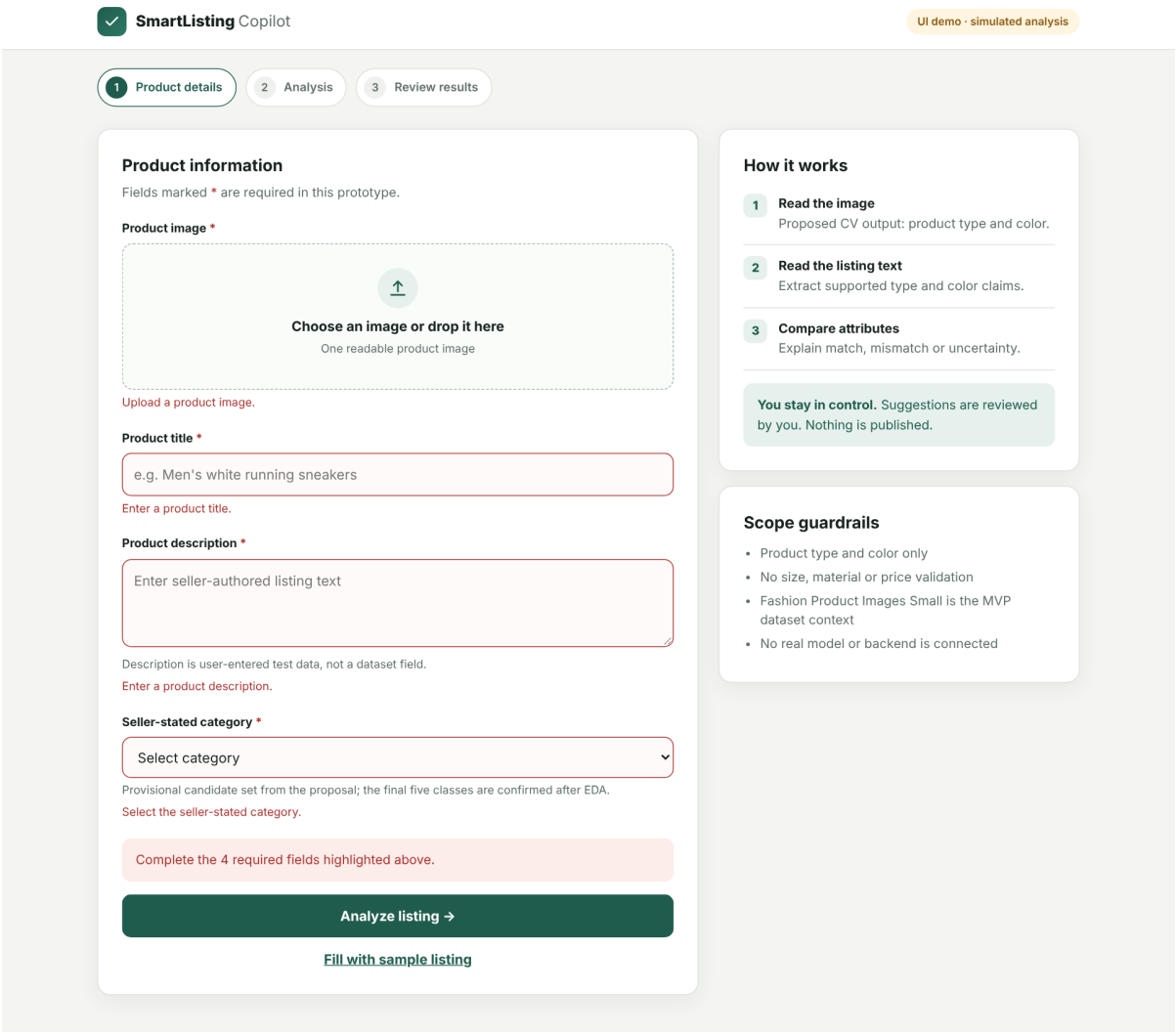



**Figure 3.** *Inline validation after submitting an empty form.*

## 6.3 Completed form

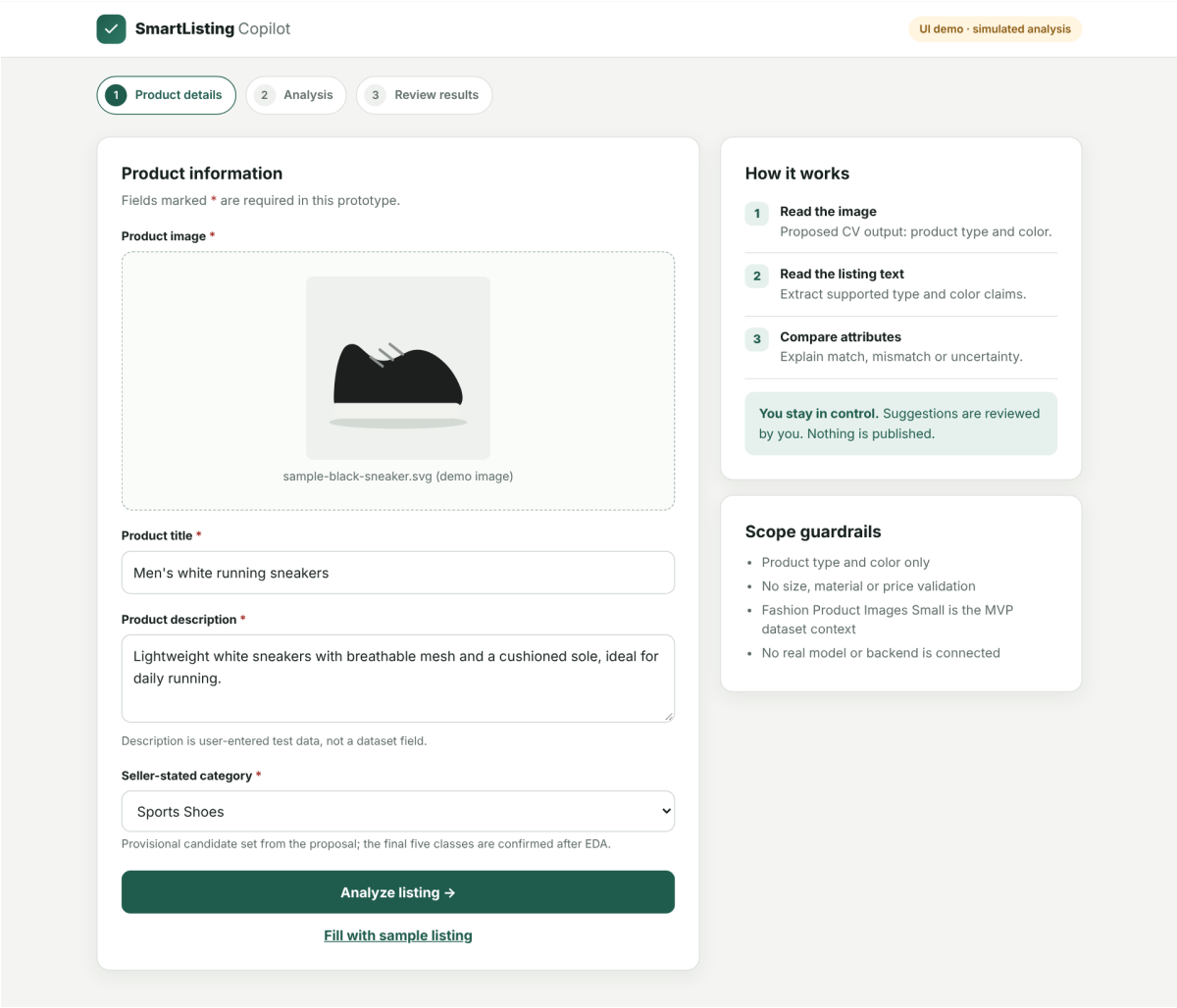



**Figure 4.** *Form completed with the sample listing and image preview.*

## 6.4 Analysis in progress

After submission the stepper moves to step 2. The screen echoes the submitted image with a scanning effect and chips for the title and category, so the user can confirm what is being analyzed. A progress bar and a four-stage pipeline show Done, In progress and Waiting in text. Duplicate submission is blocked and the demo can be cancelled, which returns to the form with entries preserved.

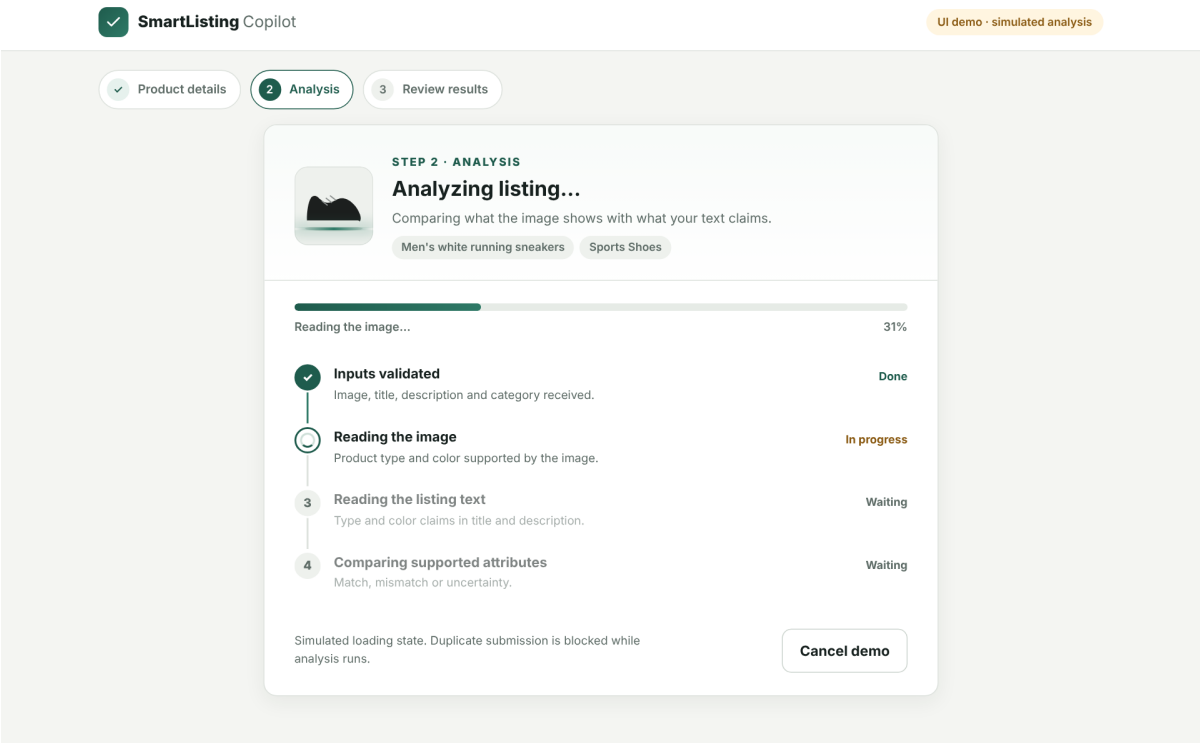



**Figure 5.** *Analysis in progress: stage 2 of 4 active, 31% complete.*

## 6.5 Review results: mismatch found

The results open with an outcome banner (icon, headline, one-line explanation and a status badge). Chips under the headline show the reason code, severity, inconsistency level and review queue defined in Section 12. Four count tiles follow. The submitted listing sits beside the attribute comparison, which separates what the image supports from what the text claims and labels each attribute with a text status, never color alone. The finding explains the contradiction. In the suggested correction, the original wording highlights the conflicting word and the suggested wording is editable beside it.

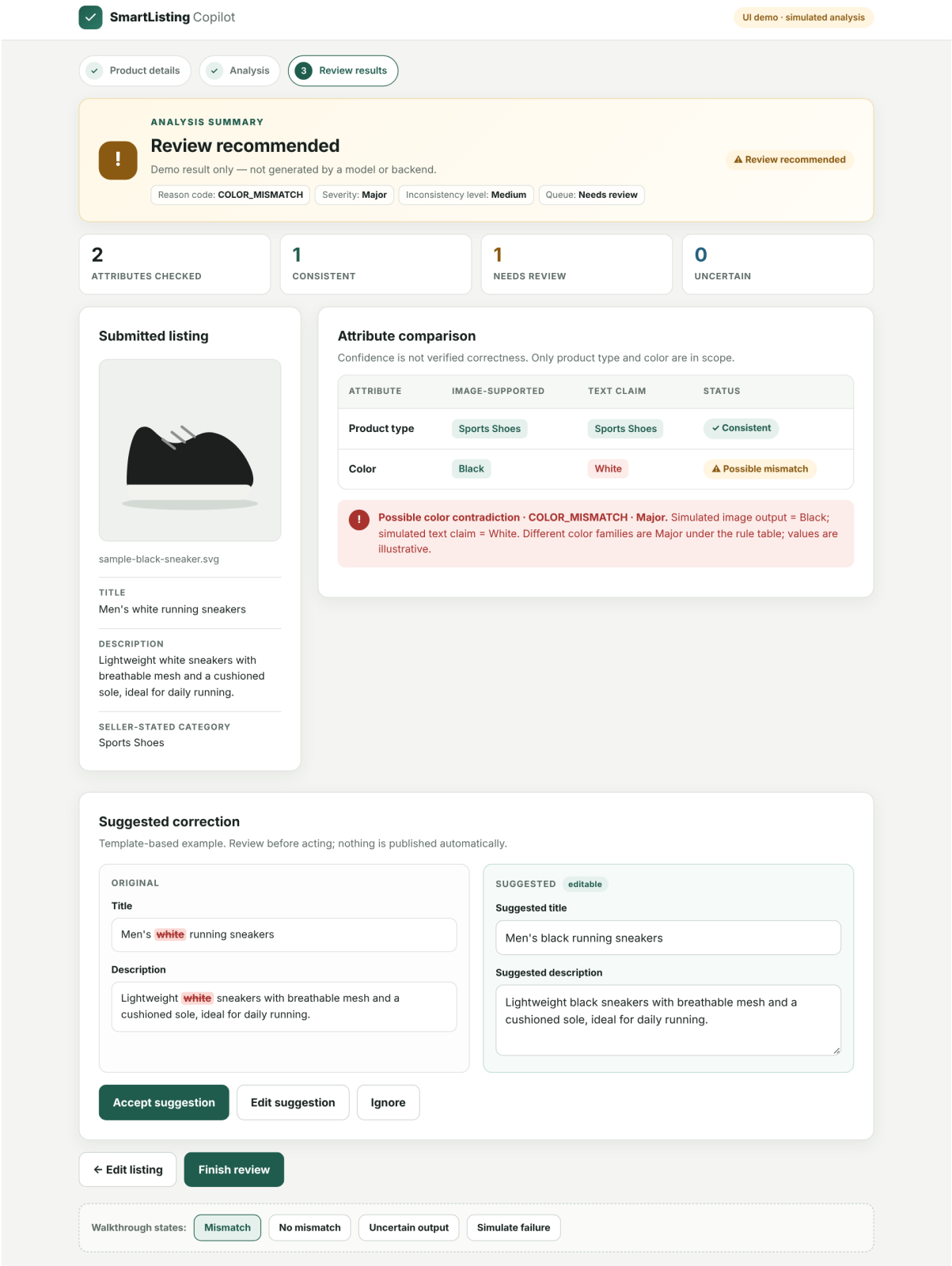



**Figure 6.** *Results with a possible color mismatch and an editable suggested correction.*

## 6.6 Results: no mismatch

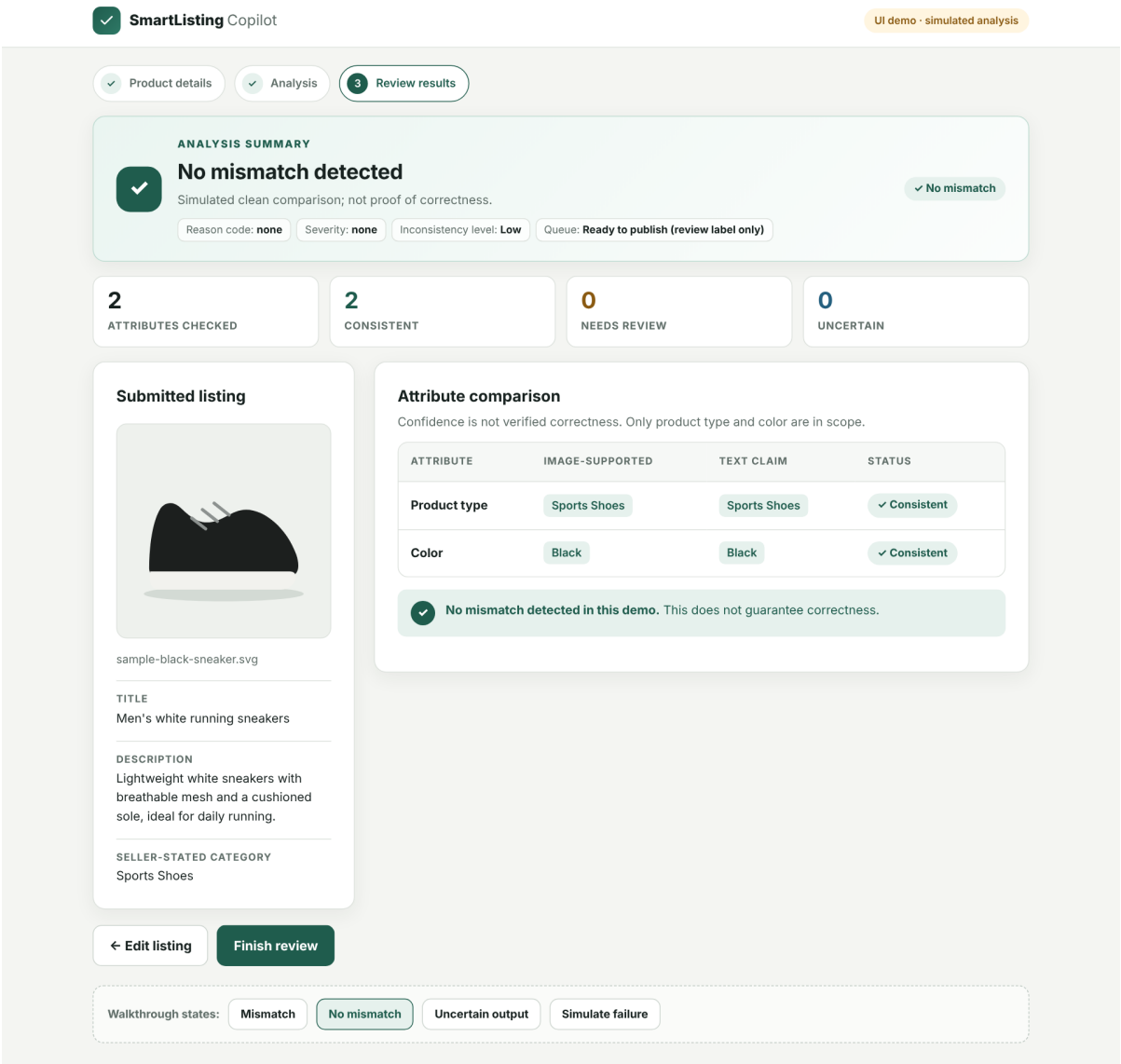



**Figure 7.** *No mismatch detected; the screen states this is not proof of correctness. The suggestion card is hidden.*

## 6.7 Results: uncertain output

When output is uncertain or unavailable, the interface does not infer a match or mismatch. The image-supported color is shown as unavailable, the suggestion actions are disabled and manual review is recommended.

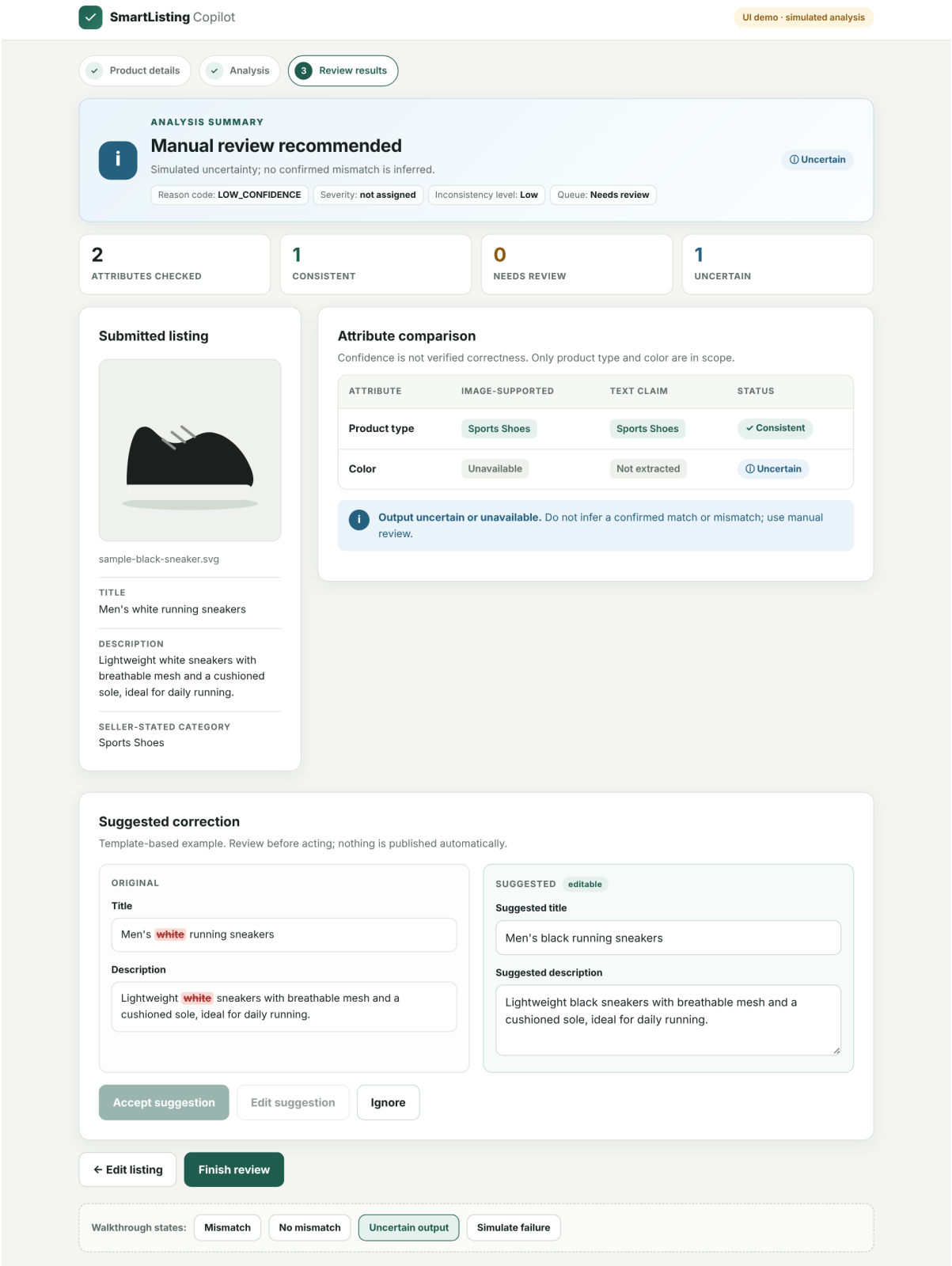



**Figure 8.** *Uncertain output state with suggestion actions disabled.*

## 6.8 Analysis failure

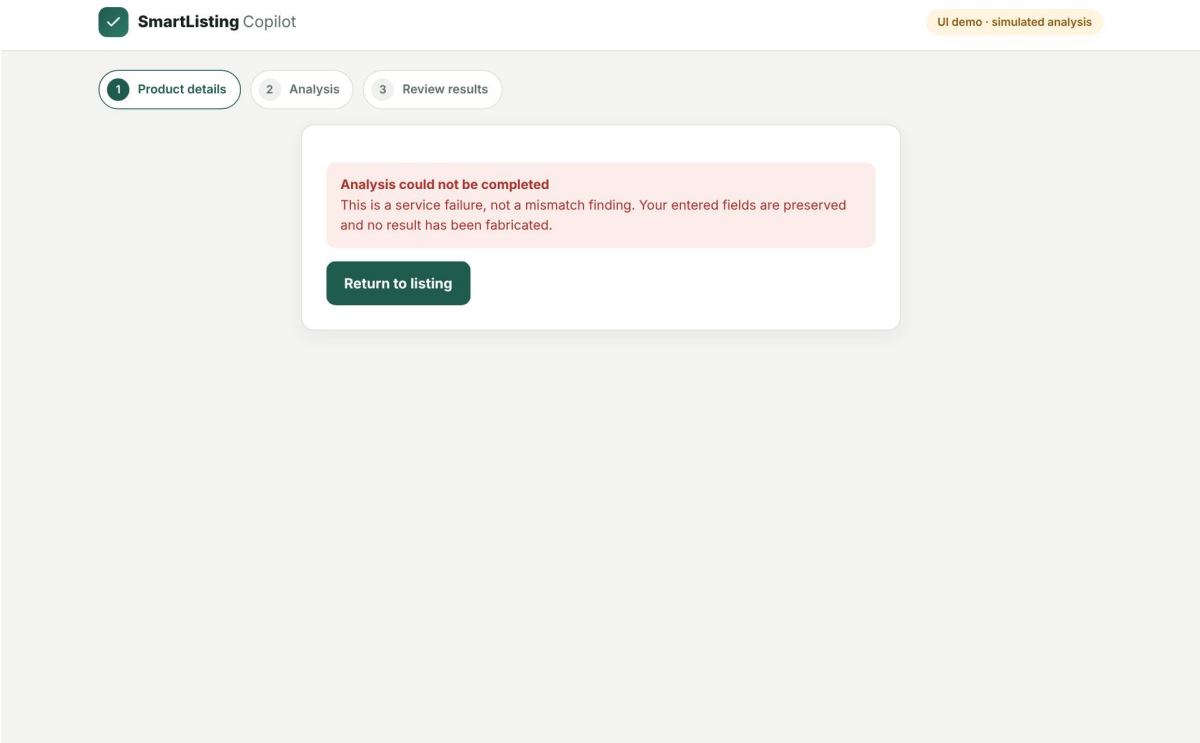



**Figure 9.** *Failure state, clearly distinguished from a mismatch finding.*

## 6.9 Responsive (mobile) layout

On narrow screens the layout collapses to a single column with full-width controls and a contained preview. The comparison table scrolls inside its own container and the count tiles reflow to two columns.

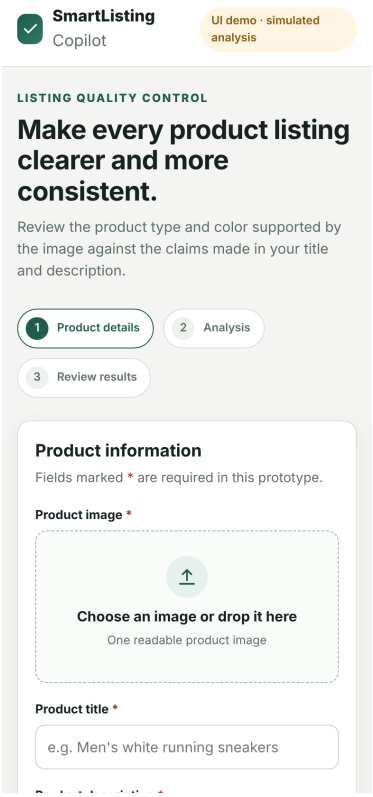



**Figure 10.** *Mobile: product input (390 px).*

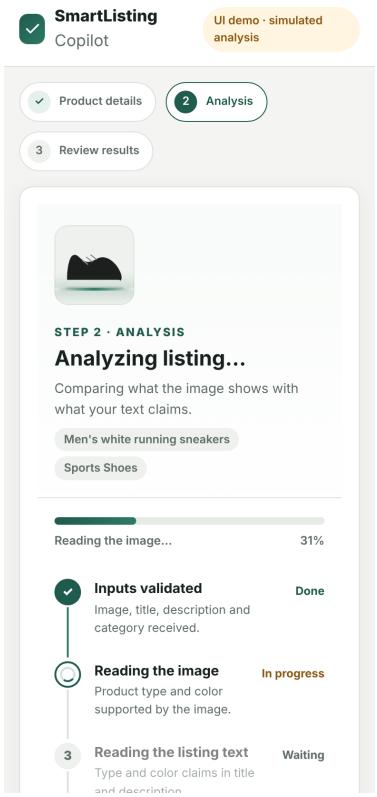



**Figure 11.** *Mobile: analysis in progress.*

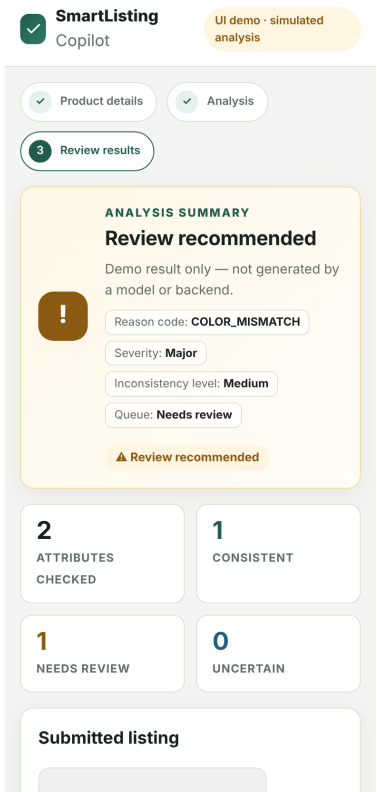



**Figure 12.** *Mobile: results (top of page).*

# 7. UI component specification

| Component | Purpose | Data | Interaction | Validation / error |
|---|---|---|---|---|
| **ImageUploader** | Select and preview the image | File, preview, filename | Choose, drag/drop, replace | Reject non-image MIME types; real readability and size checks need confirmation. |
| **ProductTitleField** | Capture title | Text | Type/edit | Required in prototype; inline error. |
| **ProductDescriptionField** | Capture seller-entered text | Text | Type/edit | Required in prototype; not a dataset field. |
| **SellerCategorySelect** | Capture declared category | Final category enum | Select | Provisional candidate set; replace with the final five classes after EDA. |
| **ReviewChips** | Show rule outcomes | Reason code, severity, inconsistency level, queue | Read | Values come from the rule tables in Section 12 only. |
| **AnalyzeButton** | Submit the form | Validated input | Submit once | Disable while pending. |
| **AnalysisProgress** | Communicate progress | Stage statuses, percentage, submitted image and chips | Cancel demo | Only truthful stages in production; progress exposed with a progressbar role. |
| **ResultsBanner** | State the outcome | Outcome, headline, explanation, badge | Read | Text and icon, not color alone; "No mismatch" is not proof of correctness. |
| **SummaryTiles** | Give counts at a glance | Checked, consistent, needs review, uncertain | Read | Counts must derive from the real comparison in production. |
| **AttributeComparison** | Compare type/color | Visual outputs, text claims, confidence/availability | Review | Distinguish match / mismatch / missing / uncertain. |
| **MismatchAlert** | Explain a finding | Attribute, values, rule, evidence | Review alert | Severity from the approved rule table only. |
| **SuggestedCorrection** | Propose a template correction | Original (conflict highlighted) and suggested text | Accept / Edit / Ignore | Only supported type/color; no auto-publish. |
| **ErrorMessage** | Explain a recoverable failure | Error category | Retry / edit / replace | Preserve input; do not imply a mismatch. |

| Component | Purpose | Data | Interaction | Validation / error |
|---|---|---|---|---|
| **UncertainOutputState** | Prevent unsupported inference | Unavailable/low-confidence values | Manual review | No fabricated claim or correction. |

## 8. Responsive design and accessibility

| Topic | Specification |
|---|---|
| **Desktop** | Centered max-width layout; two-column form and guidance; results split into listing and comparison, with a full-width suggestion card. |
| **Mobile** | Single column; full-width controls; contained preview; readable stacked comparisons. |
| **Keyboard** | Logical tab order, keyboard-accessible uploader, visible focus ring. |
| **Labels / errors** | Persistent labels, textual required state, field-level errors connected with aria-describedby; submission errors announced with role="alert". |
| **Contrast / status** | Neutral surfaces with a restrained green accent; status text and icons in addition to color. |
| **Loading** | Text status, progress bar with a progressbar role and a live region; disable duplicate submit; honor the reduced-motion preference. |
| **Images** | Meaningful alt text for the uploaded preview and the submitted image. |
| **Testing boundary** | No formal WCAG audit or certification has been performed; keyboard, screen-reader, contrast and mobile testing remain. |

## 9. UI design rationale

- **Single-task focus** keeps the interface centered on listing review rather than dashboard widgets.
- **Evidence-first comparison** separates image-supported output from text claims.
- **Outcome before detail:** the banner and count tiles answer "what happened?" before the user reads the evidence.
- **Transparent uncertainty** avoids treating unavailable output as a confirmed match.
- **Before/after suggestions** with the changed word highlighted preserve seller control and never publish automatically.
- **Visible progress** with truthful stage names reduces perceived waiting and shows what is being compared.
- **Required markers, preview and inline errors** prevent common input mistakes.
- **Consistent stepper, terminology and return-to-edit actions** orient the user.
- **Scope discipline:** only product type and color appear in comparisons; unsupported attributes are not guessed.
- **Restrained visual styling** gives evidence and warnings priority.

# 10. UI-to-requirements traceability

| Reference | Intent | UI location | Evidence / status |
|---|---|---|---|
| **FR-001 to FR-006** | Image, text and category input; visual type, color and image-quality outputs | Input, preview, results | IDs verified against the FRS. Upload works; model outputs simulated. |
| **FR-007 to FR-008** | Extract supported type/color claims; English/Arabic-script route | Fields and comparison | No real NLP or Arabic routing connected. |
| **FR-009** | Compare image-supported attributes with text claims | Comparison table | Simulated only. |
| **FR-010** | Type mismatch and severity rules | Finding card | Rule table must drive production severity. |
| **FR-011** | Color-family and near-shade rules | Finding card | Illustrative mismatch only. |
| **FR-012** | Detect absent supported claims | Comparison / missing state | Design supports "Not stated"; live logic absent. |
| **FR-013** | Expose confidence without implying correctness | Results / uncertainty | Copy included; values simulated. |
| **FR-021 to FR-025; US-07/US-08 mapping** | Before/after suggestion and Accept/Edit/Ignore | Suggestion card | Local demo actions only. |
| **NFR-USA-01/02/04/05/06 (confirm baseline)** | Usability, clear warnings, user control | Form and results | Walkthrough testing pending. |
| **NFR-REL-01** | Safe handling of blurred or out-of-scope images | Upload / error states | Error layout shown; real image-quality checks absent. |

**Traceability note.** FR, US and AC identifiers were verified against the FRS (16 October 2026). NFR identifiers come from a separate non-functional requirements document that was not provided; confirm them before submission.

# 11. Open questions and dependencies

| Open item | Why it matters | Decision owner |
|---|---|---|
| **Five EDA-selected categories** | Prototype uses the proposal's illustrative set (Section 12.1); it must match the final dataset audit. | EDA/CV owner freezes the enum. |
| **Accepted image encodings and max size** | Not specified in the FRS or proposal; Section 12.2 treats them as configurable constants. | Backend/integration owner. |
| **Empty title/description behavior** | The FRS notes reject-vs-missing behavior as unresolved. | Requirements and NLP owners. |

| Open item | Why it matters | Decision owner |
|---|---|---|
| **Seller category mapping** | Must align with the final classes and comparison rules. | Requirements/CV/NLP owners. |
| **Confidence threshold and score weights** | Severity bands and queue rules are defined (Section 12.3 to 12.5); the numeric threshold comes from validation data and the weights are deferred. | Integration/classifier owners. |
| **Suggestion templates** | Must only correct supported attributes and preserve control. | NLP/integration owners. |
| **Arabic UI vs text processing** | Script detection does not automatically imply full UI localization. | Team decision. |
| **API contract and persistence** | Contract proposed in Section 13; the prototype is not connected to services. | Backend/deployment owner. |
| **Accessibility validation** | No formal audit has been performed. | Team testing and sign-off. |

# 12. Category set and validation rules

This section records the rules the interface must follow. Each row states its source and whether the source fixes the value. Values that the FRS and the proposal defer are marked **Open**; they are not invented here.

## 12.1 Category set

| Item | Value | Status |
|---|---|---|
| **Number of classes** | Exactly five product types (proposal 6.1; FRS section 12). | Frozen by source |
| **Data source** | Fashion Product Images Small only. Type comes from articleType and color from baseColour. | Frozen by source |
| **Selection criteria** | Class balance, image availability and label quality, assessed in EDA (proposal section 10). | Frozen by source |
| **Candidate classes** | Casual Shoes, Formal Shoes, Sports Shoes, Shirts, Handbags. The proposal gives these as examples and says they are illustrative, not the final list. | Provisional |
| **Freeze step** | The EDA and CV owners confirm the five labels, the type groups (from subCategory), the color-family normalization and the near-shade tolerance. The UI dropdown and the rule tables then read from one shared list. | Pending EDA |
| **Where it appears** | Seller-stated category dropdown; type column of the comparison; unsupported-category check (FR-015). | Prototype uses the provisional set |

## 12.2 Input validation rules

| Input | Rule | Source | Status |
|---|---|---|---|
| **Product image** | Required. One image. Must be readable for analysis. A missing image does not start analysis. | FR-001, AC-001, FRS section 7 | Frozen by source |
| **Image format and size** | No accepted encodings or size limit are specified. Keep them as configurable constants owned by the backend. | FRS section 7 note | Open |
| **Product title** | Required. Passed to NLP unchanged. Unknown vocabulary yields no inferred claim. | FR-002, FR-007 | Frozen by source |
| **Product description** | Required. Seller-entered or synthetic text. It is not a dataset field. | FR-002, FRS section 7 | Frozen by source |
| **Seller category** | Required. Compared with the final category set and the visual signals. A conflict is reviewed. | FR-003, FR-015 | Frozen by source |

| Input | Rule | Source | Status |
|---|---|---|---|
| **Empty title or description** | The source leaves open whether this is rejected or processed as a missing claim. The prototype rejects it at submission with an inline error. This is a prototype decision awaiting confirmation. | FRS section 7 note | Open |
| **Text length limits** | Not specified. | FRS section 7 note | Open |
| **Language route** | Arabic characters select the fixed Arabic lexicon; all other text uses the English vocabulary. | FR-008 | Frozen by source |
| **Error wording** | Inline, next to the field, in plain language. A service failure is never worded as a mismatch. | Design decision | Proposed |

## 12.3 Findings, severity and reason codes

| Condition | Example | Severity / outcome | Reason code | Source |
|---|---|---|---|---|
| **Type, different group** | Image: shirt. Title: shoes. | Critical | TYPE_MISMATCH | FR-010, FR-017 |
| **Type, same group** | Image: formal shoe. Title: sports shoe. | Major | TYPE_MISMATCH | FR-010, FR-017 |
| **Color, different family** | Image: brown. Title: black. | Major | COLOR_MISMATCH | FR-011, FR-017 |
| **Color, near shade** | Image: navy. Title: blue. | Minor or ignored | COLOR_MISMATCH | FR-011, FR-017 |
| **Low model confidence** | Prediction below the validation-derived threshold | Needs review | LOW_CONFIDENCE | FR-014 |
| **Blurred image** | Blur rule is triggered | Needs review | IMAGE_BLUR | FR-006 |
| **Missing attribute** | Image shows a color the text omits | Needs review | MISSING_ATTRIBUTE | FR-012 (Should) |
| **Unsupported category** | CV and CLIP disagree, the category contradicts the seller, or CLIP prefers "other product" | Needs review | UNSUPPORTED_CATEGORY | FR-015 |

Every surfaced warning carries one reason code and a plain-language explanation (FR-016). Confidence is shown beside it and is never described as proof of correctness (FR-013).

## 12.4 Listing inconsistency level

| Signals detected | Level |
|---|---|
| **No Major or Critical mismatch** | Low |
| **One Major mismatch** | Medium |
| **Two or more Major mismatches** | High |
| **Any Critical mismatch** | High |

Low confidence and image blur do not change the level. They route the listing to Needs review through the queue rules (FR-019). The level is a heuristic about listing consistency and is never presented as a return probability.

## 12.5 Review queue

| Queue | Rule | Prototype state |
|---|---|---|
| **Ready to publish** | Low inconsistency level and no low-confidence, image-blur or missing-attribute flag. This is a review label only; nothing is published. | No mismatch |
| **Needs review** | Minor mismatch, low confidence, image blur, missing attribute or unsupported category. | Uncertain output; the Major color demo with a Medium level |
| **Critical mismatch** | High inconsistency level or any Critical mismatch. | Not shown in the prototype |

The source assigns a single Major mismatch a Medium level, so the demo mismatch lands in Needs review. Handling of ties and combined flags is not specified.

## 12.6 Values still deferred by the source

| Value | Why it is deferred | Owner |
|---|---|---|
| **Numeric confidence threshold** | Chosen on validation data after baselines and before final test evaluation. | CV / integration |
| **Listing Quality Score weights** | Three components are fixed (image quality, attribute completeness, image-text consistency); weights are not. | Integration |
| **Color families and near-shade tolerance** | Fixed after EDA and color normalization. | CV / NLP |
| **Tie and combined-flag handling in the queue** | Not specified. | Integration |

# 13. Pipeline integration contract

The FRS and proposal describe a FastAPI service that runs a CV branch and an NLP branch in parallel, merges them at comparison, then scores, routes and generates the suggestion. A Streamlit interface displays the result. The HTML prototype is the design reference for those screens. This section defines what the interface needs from the service so real modules can replace stubs without changing the screens.

> **Status.** The contract below is specified. The real CV, NLP, comparison and scoring modules are not connected; the prototype still uses hard-coded scenarios. Connecting them is implementation work outside this design package.

## 13.1 Unified data contract

The proposal fixes the stages and a few field names (type, color, confidence, quality, claimed_type, claimed_color, language). The remaining field names in this section are proposed.

| Stage | Fields | Used by the interface for |
|---|---|---|
| **Input** | image, title, description, category | The form and the submitted-listing card. |
| **CV output** | type, color, confidence, quality (resolution, blur, brightness) | The image-supported column, confidence note and IMAGE_BLUR. |
| **NLP output** | claimed_type, claimed_color, language | The text-claim column and the language route. |
| **Comparison** | mismatches, missing attributes, reason codes, severity, explanation | Status badges, the finding and the review chips. |
| **Scoring** | quality score, inconsistency level, queue | Review chips and the count tiles. |
| **Final response** | All of the above plus suggested title and description | The suggestion card and the banner. |

## 13.2 Example response (illustrative)

This is the demo mismatch scenario written in the proposed shape. The numbers are placeholders, not model output.

```
{
  "analysis_status": "ok",
  "cv": {"type": "Sports Shoes", "color": "Black",
         "confidence": {"type": 0.94, "color": 0.94},
         "quality": {"resolution_ok": true, "blur": false, "brightness_ok": true}},
  "nlp": {"language": "en", "claimed_type": "Sports Shoes", "claimed_color": "White"},
  "comparison": {
    "mismatches": [{"attribute": "color", "image": "Black", "text": "White",
                    "severity": "Major", "reason_code": "COLOR_MISMATCH",
                    "explanation": "Different color families."}],
    "missing_attributes": []},
  "scoring": {"quality_score": null, "inconsistency_level": "Medium",
              "queue": "Needs review"},
  "suggestion": {"title": "Men's black running sneakers",
                 "description": "Lightweight black sneakers with breathable mesh..."}
}
```

## 13.3 Interface-to-contract mapping

| Interface element | Contract field | Rule |
|---|---|---|
| **Image-supported column** | cv.type, cv.color | Show "Unavailable" when null or low confidence. |
| **Text-claim column** | nlp.claimed_type, nlp.claimed_color | Show "Not stated" when absent (FR-012). |
| **Status badge** | comparison.mismatches, missing_attributes | Consistent, Possible mismatch, Not stated or Uncertain. |
| **Banner headline and tone** | scoring.queue, analysis_status | Section 12.5 decides the wording. |
| **Reason code and severity chips** | comparison.mismatches[].reason_code, severity | Only codes from Section 12.3. |
| **Inconsistency and queue chips** | scoring.inconsistency_level, scoring.queue | Only values from Section 12.4 and 12.5. |
| **Count tiles** | Derived from the comparison | Never typed in by hand. |
| **Suggestion card** | suggestion.title, suggestion.description | Hidden when no supported attribute is wrong (FR-021). |
| **Accept, Edit, Ignore** | Stays on the client | No publish call exists (FR-023 to FR-025). |

## 13.4 Error contract

| Situation | Signal | Interface response |
|---|---|---|
| **Missing or invalid input** | Caught before the request | Inline field errors; nothing is sent. |
| **Service failure or timeout** | Non-success response or no response | Failure screen; fields preserved; no result invented. |
| **Output uncertain or unavailable** | Null fields or confidence below the threshold | Uncertain state; suggestion actions disabled. |
| **Unsupported category** | Reason code UNSUPPORTED_CATEGORY | Needs review without a forced type. |

## 13.5 Integration steps

The proposal calls for a stub-first approach: one shared schema, each module first returning schema-valid dummy output, then real modules replacing stubs one at a time. Build order: schema and stubs, dataset, CV, NLP, mismatch engine, scoring, generation, API, UI. Docker is added once the pipeline is stable.

| Step | Status | Owner (provisional split) |
|---|---|---|
| **Interface fields and states defined (Sections 12 and 13)** | Done | Member 5 |
| **Shared schema and stubs** | Pending | Integration |

| Step | Status | Owner (provisional split) |
|---|---|---|
| **Replace the demo scenarios with a call to the service** | Pending | API / UI |
| **Real CV module** | Pending | CV |
| **Real NLP module** | Pending | NLP |
| **Mismatch engine and scoring** | Pending | Integration |
| **End-to-end acceptance run against AC-001 to AC-025** | Pending | Whole team |

# 14. Stakeholder walkthrough and sign-off pack

> **Status.** This pack is prepared. The walkthrough has not been held, and no approval is recorded in this document. The sign-off table is intentionally blank.

## 14.1 Participants and aim

The FRS names two operational users and one development stakeholder. The aim is to confirm that the seller and the reviewer can read the result correctly, and that the team agrees on the open decisions in Section 11.

| Participant | Role in the FRS | Looks at |
|---|---|---|
| **Marketplace seller (or proxy)** | Primary user | Input, analysis, suggestion and control over changes. |
| **Catalog reviewer (or proxy)** | Review user | Reason codes, severity, queue and explanations. |
| **Development team** | Verification stakeholder | Contract, open decisions and feasibility. |

## 14.2 Walkthrough script

| # | Action in the prototype | What to observe | Question to ask |
|---|---|---|---|
| 1 | Open prototype.html | Scope message and stepper | Is it clear what the tool checks? |
| 2 | Choose Analyze listing with an empty form | Inline errors; focus moves to the first invalid field | Is it clear what to fix? |
| 3 | Choose Fill with sample listing | Image preview and filled fields | Is the preview what you expect? |
| 4 | Choose Analyze listing | Staged progress, submitted image, Cancel demo | Is it clear what is being compared? |
| 5 | Read the mismatch result | Banner, chips, comparison table, finding | Can you say what the image shows and what the text claims? |
| 6 | Use Accept, Edit and Ignore | Highlighted word; the note that nothing is published | Do you feel in control of the final wording? |
| 7 | Choose No mismatch | Statement that this is not proof of correctness | Does the wording avoid overconfidence? |
| 8 | Choose Uncertain output | Disabled suggestion actions; Needs review | Is it clear why no correction is offered? |
| 9 | Choose Simulate failure | Failure message distinct from a mismatch | Could this be mistaken for a mismatch? |
| 10 | Narrow the window to phone width | Single column; scrolling comparison table | Is everything still readable? |

## 14.3 Success criteria

| Criterion | Source | Result |
|---|---|---|
| **The user states what the image supports and what the text claims.** | US-04, FR-009 | Pass / Fail |
| **Each warning shows a reason code and an explanation.** | FR-016, AC-016 | Pass / Fail |
| **Confidence is not read as proof of correctness.** | FR-013, AC-013 | Pass / Fail |
| **The user can tell original text from suggested text and sees what changed.** | FR-022, AC-022 | Pass / Fail |
| **The user understands that Accept, Edit and Ignore publish nothing.** | FR-023 to FR-025 | Pass / Fail |
| **An uncertain result does not claim a match or a mismatch.** | FR-014, FR-015 | Pass / Fail |
| **A service failure is not mistaken for a mismatch.** | Design rule | Pass / Fail |

## 14.4 Feedback log

| Date | Participant role | Observation | Severity | Decision / owner |
|---|---|---|---|---|
| | | | | |
| | | | | |
| | | | | |

## 14.5 Decision and sign-off

Each participant records one decision after the walkthrough: Approve, Approve with changes, or Reject.

| Name | Role | Decision | Date | Signature |
|---|---|---|---|---|
| | | | | |
| | | | | |
| | | | | |

# Completion checklist

- ☑ Scope and assumptions
- ☑ Persona and journey
- ☑ User flow diagram (image and Mermaid source)
- ☑ Low-fidelity screen and state specifications
- ☑ Interactive standalone prototype with screenshots
- ☑ Component specification
- ☑ Responsive/accessibility guidance
- ☑ Design rationale and traceability
- ☑ Open dependencies and approval decisions
- ☑ Category set and exact validation rules specified (Section 12; category list provisional until EDA)
- ☑ Pipeline integration contract specified (Section 13; real modules not yet connected)
- ☑ Stakeholder walkthrough and sign-off pack prepared (Section 14)
- ☐ Hold the walkthrough and collect signatures
- ☐ Confirm the final five categories after EDA
- ☐ Connect the real CV, NLP and scoring modules

# How to open and present

- Open prototype.html in Chrome, Edge or Firefox; no installation is needed.
- Click "Fill with sample listing", or upload any image and complete the required fields, then choose "Analyze listing" to see the simulated flow.
- Use the walkthrough buttons on the results screen to show the mismatch, no-mismatch, uncertain-output and failure states.
- Open user_flow.mmd in a Mermaid-compatible editor to re-render or edit the flow diagram.

*End of package. Version 1.2, prepared for team review.*